

ERAM RAHMAN

US Citizen | San Diego, CA (UC San Diego) | Los Angeles, CA | Open to Relocation |
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EDUCATION

Expected Graduation: 2027

Bachelor of Science in Computer Engineering — University of California San Diego

Major: Computer Engineering

Relevant Coursework: Data structures and Algorithms, Software Engineering, Digital Systems, Computer Architecture, Operating Systems

SKILLS

- **Programming Languages:** Python, C++, C, Assembly (ARM), HTML, CSS, Java, JavaScript, MATLAB
- **Web and Application Development:** React, Next.js, JavaFX, SQL, Git/GitHub, REST APIs
- **IT and Hardware:** PC Assembly, Hardware Troubleshooting, System Diagnostics

EXPERIENCE

Undergraduate Researcher — Robot Perception & Mapping

January 2026 - Present

- Implemented path planning algorithms (A*, RRT) in simulation to evaluate robot navigation and decision-making
- Developed a robotic arm simulation to test motion planning and pathfinding algorithms
- Led development efforts for a research workshop, coordinating tasks and guiding team members to complete a technical paper and presentation on robot perception and planning, increasing efficiency by 30%

PC Technician and Owner

February 2019 - Present

- Provided technical consulting and support to 100+ clients, diagnosing hardware and software issues with a high success rate
- Reduced client downtime by 30–50% through efficient troubleshooting and hardware optimization
- Provided technical support and hardware recommendations

GiosTech Internship (In Progress, Completing June 2026)

- Developed an interactive web application using Three.js to render and interact with 3D human body models
- Implemented AI-driven logic to generate personalized exercise recommendations based on user-reported pain locations
- Built frontend and backend functionality to process user input and deliver dynamic recommendations

PROJECTS

PC Recommendation Website

- Developed a full-stack web application using Next.js (React), achieving perfect Lighthouse scores (100/100/100) with sub-second load times (FCP ~0.3s, LCP ~0.4s)
- Implemented a questionnaire-based system to match users with optimal builds based on budget and use case
- Integrated external APIs to fetch product data and implemented server-side filtering on ~20 products per request based on user survey inputs

Voice Controlled Navigation

- Built a web-based navigation application using Mapbox API with real-time GPS tracking and routing, achieving ~150–170 ms API response times and a Lighthouse performance score of 100
- Implemented voice command functionality for hands-free navigation, reducing manual interaction by ~25–35% and improving user safety
- Developed dynamic route recalculation and multi-stop navigation, improving routing efficiency and responsiveness

Student Clock Application

- Developed a Java-based productivity application using JavaFX and SQLite for task and time management
- Implemented priority-based scheduling algorithm to optimize study time allocation by ~50%
- Designed an MVC architecture with persistent data storage and real-time tracking features